University of Florida, Department of Mathematics

Topology First-Year Exam, May 2021, Part 1

Part 1.

Work each of the following problems and show all work. Support all statements to the best of your ability.

1. Let S be a set, and let $\mathcal{P}(S)$ denote the set of subsets of S. Prove that there is no surjective function $f : S \to \mathcal{P}(S)$.

2. Let X and Y be connected topological spaces. Prove that $X \times Y$ is connected, where $X \times Y$ has the product topology.

3. Prove that every compact, metrizable space has a countable basis.

4. Prove that the one-point compactification of $\mathbb{Z}_+$ is homeomorphic to the subspace $\{0\} \cup \left\{\frac{1}{n} : n \in \mathbb{Z}_+\right\}$ of $\mathbb{R}$.

5. Recall that $\mathbb{R}_K$ denotes the real line with the K-topology, and the set K is given by $K = \left\{\frac{1}{n} : n \in \mathbb{Z}_+\right\}$. Let Y denote the quotient space obtained from $\mathbb{R}_K$ by collapsing the set K to a point. Prove that Y satisfies the T_1 axiom, but is not Hausdorff.

Part 2.

Answer the following with complete definitions or statements or short proofs.

6. In the finite complement topology on $\mathbb{R}$, to what point or points does the sequence $x_n = \frac{1}{n}$ converge?

7. Is every simply ordered set a Hausdorff space in the order topology?

8. Define compact, limit point compact, and sequentially compact.

9. If $f : X \to Y$ is a continuous surjective function, and X is path connected, must Y be path connected?

10. State the Intermediate Value Theorem.